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The tech advantage: exploring technological determinants of social media marketing adoption in Pakistani small and medium startups

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Abstract

This study investigates the technological determinants influencing social media marketing (SMM) adoption among Pakistani Small and Medium Startups (SMS), with a focus on the mediating role of technology knowledge. Utilizing the technology–organization–environment (TOE) Framework, the research examines the impact of technology compatibility, complexity, cost, infrastructure, and knowledge on the adoption of SMM. Data were collected through a survey from 386 startups based in Karachi, providing a robust basis for analysis. The findings reveal that technology knowledge significantly mediates the relationships between these technological factors and SMM adoption, enabling startups to better navigate the complexities associated with SMM. Results show that technology compatibility, infrastructure, and knowledge positively influence SMM adoption, while technology complexity and cost are not significant barriers for startups. This contrasts with traditional SMEs, where these factors often hinder SMM adoption efforts. The study highlights the distinct nature of SMSs and the necessity of tailoring technology adoption strategies to their unique needs. The practical implications of this research highlight the importance of investing in technology training and infrastructure to empower startups with the necessary tools and skills for leveraging SMM effectively. This approach not only enhances their market competitiveness but also supports their growth in the digital economy. Overall, this research contributes to a deeper understanding of how technological determinants affect SMM adoption in startups, providing valuable insights for both practitioners and policymakers aiming to promote a robust digital startup ecosystem.

Keywords: Technology, Small and medium enterprises, Small and medium startups, Social media marketing, TOE framework

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Introduction

Businesses across the globe are increasingly leveraging the power of social media marketing to enhance their visibility, engage with customers, and drive growth (Khanom, 2023). This shift towards digital platforms is part of a broader technological revolution that was initiated through enterprise resource planning systems, cloud computing and data analytics. The emergence of these technologies has significantly improved manageability, agility, efficiency, and profitability of the businesses. Social media marketing, in particular, offers a unique opportunity for businesses to create, communicate, and deliver value directly to their customers in real time, promoting a more dynamic and interactive customer relationship (Nuseir et al., 2023; Purnomo, 2023).

Pakistan is home to over 5 million SMEs, constituting 90% of all private enterprises. These SMEs play a pivotal role in the economy by contributing about 30% to the Gross Domestic Product (GDP) and generating nearly 78% of non-agricultural employment (Khowaja et al., 2021; Najeeb, 2021). In addition, SMEs are responsible for creating 90% of new jobs in the formal sector, signifying their importance in addressing unemployment (Enaifoghe & Vezi-Magigaba, 2023). Given that a 1% increase in output of SMEs leads to a 1.7% increase in GDP (Manzoor et al., 2021), it becomes evident that supporting and increasing the number of Small and Medium Startups (SMS) is essential for economic growth and prosperity.

Small and Medium Enterprise Development Authority (SMEDA) Pakistan, defines SMS as Small and Medium Enterprises that are less than 5 years (SMEDA, 2022). These SMSs, as identified by SMEDA faces significant challenges and a failure rate as high as 90% (Kyril Kotashev, 2022;Sen, 2023;Zaidi et al., 2023). A primary factor contributing to this high failure rate is ineffective and inadequate marketing strategies, combined with financial constraints that make traditional marketing platforms inaccessible for these startups (Wilson & Makau, 2017). While social media marketing presents a cost-effective alternative, the literature identified its adoption among established SMEs as slow, largely due to technological barriers and lack of digital literacy (Teixeira et al., 2018; Verma et al., 2023), but the study related to technological issues faced by SMS in social media marketing adoption provides a significant gap.

Social media marketing adoption is significantly hindered by technological barriers, due to complexity of digital technologies and the fast-paced evolution of online platforms (Bradač Hojnik & Huđek, 2023). For businesses that are lacking in expertise or resources, the task of navigating and effectively utilizing social media marketing strategies could be challenging (Braspenning et al., 2022; Mabad et al., 2021). In addition, the challenge of keeping up with the latest technological advancements and understanding how to measure the return on investment for social media initiatives adds another layer of difficulty to the adoption process (Rahman et al., 2021). Despite the development of cost-effective and user-friendly social media tools that are designed to ease the adoption process, the gap in technology knowledge remains a prominent obstacle (Emini & Merovci, 2021). This knowledge gap acts as a mediating variable, where firms with a deeper understanding of digital tools are in a better position to leverage social media marketing effectively, highlighting the importance of enhancing technology knowledge to overcome these barriers (Prasetya & Susilo, 2022). Given the discussion above, the research seeks to answer following questions:

1. What are the technological determinants of Social Media Marketing Adoption in Pakistani SMS?
2. How does Technology Knowledge mediate the relationship between technological determinants and Social Media Marketing Adoption?

For addressing the technological factors involved in social media marketing adoption, the research will follow technology–organization–environment (TOE) framework, thus expanding the body of knowledge by linking it with SMS by examining relevant technological determinants for social media marketing adoption. The current study has theoretical and practical significance in the field of marketing, entrepreneurship and innovation. Although, social media marketing adoption has received attention from researchers but specific adoption pattern and technological factors related to SMSs still offers a significant gap to be studied and explored.

Literature and framework development

Small and medium startups and social media marketing adoption

The advent of technologies such as social media marketing (SMM) has transformed the way businesses engage with their audience and market their products or services. Utilizing platforms such as Facebook, Instagram, WhatsApp, YouTube, and LinkedIn, SMM offers an affordable and effective platform for businesses, especially SMS, to promote themselves. This technological advancement not only facilitates the dissemination of firm-related content but also promotes a collaborative environment, where stakeholders, including customers, can actively participate in content creation, thereby enhancing the business's visibility and reach (Cvjetković, 2023; Isohella et al., 2017; Mittal et al., 2023). Despite the clear benefits, the SMM adoption in developing countries has been slow, mainly attributed to a lack of understanding of its value and potential advantages for business growth and customer engagement (Pollák & Markovič, 2021; Radicic & Petković, 2023).

The significance of SMM for SMSs is indisputable. As highlighted by various studies, SMM's cost-effectiveness and its ability to reach a wider audience without geographical limitations make it an essential tool for new businesses striving for brand recognition and consumer base expansion (Eduardsen et al., 2023; OECD, 2021). Moreover, SMM enables businesses to measure the impact of their marketing campaigns in real time through engagement metrics, such as views, clicks, and shares, offering insights into consumer behavior and preferences. This measurable feedback function not only allows for the optimization of marketing strategies but also significantly improves the return on investment, making SMM a critical factor in the survival and success of SMS in the competitive digital marketplace (Royle & Laing, 2014; Yasmin et al., 2015).

Technology–organization–environment (TOE) framework

The technology–organization–environment (TOE) framework, proposed by Tornatzky and Fleischer (1990), provides a comprehensive model for understanding the factors that influence the adoption of technological innovations within organizations. This framework categorizes the determinants of technology adoption into three contexts: technological, organizational, and environmental. In our study, we specifically focused on the

technological context of the TOE framework to explore the factors influencing social media marketing adoption among Pakistani SMS. The technological context includes variables, such as technology compatibility, technology complexity, technology cost, technology infrastructure and technology knowledge. These factors were chosen based on their relevance and prominence in previous studies on technology adoption (Zhu et al., 2002; Pan & Jang, 2008; Effendi et al., 2020; Almansour, 2022). The TOE framework guided us in selecting these determinants by emphasizing the importance of evaluating both the existing technologies and those available for adoption, as well as understanding how these technologies align with the operational practices and resources of the firm. We chose to focus exclusively on technological determinants for several reasons. First, the rapid evolution of social media marketing technologies and their critical role in the competitive landscape for startups necessitate a deep understanding of the specific technological challenges and opportunities. Second, while organizational and environmental factors are also important, the unique technological barriers and enablers present more immediate and tangible obstacles face by SMS in social media marketing adoption. By narrowing our focus to technological determinants, we aim to provide targeted insights and practical recommendations that directly address these critical aspects. This systematic approach allowed us to develop a focused framework for analyzing the impact of technological factors on social media marketing adoption, ensuring that our study addresses the key elements that influence technology adoption decisions within SMS.

Technology compatibility and social media marketing adoption

The concept of technology compatibility, as defined by Rogers (1995) and further refined by following researchers, plays a crucial role in the social media marketing adoption within startups. Compatibility, defined as the degree to which social media marketing aligns with existing values, practices, and needs of startups, significantly influences the decision to adopt these platforms for marketing purposes. When startups perceive social media technologies as consistent with their operational style, cultural values, and business models, they are more likely to embrace these tools, leading to higher diffusion rates and more successful outcomes (Eze et al., 2021; Sombultawee, 2020). This alignment reduces uncertainties and risks associated with new technology adoption, facilitating a smoother integration into existing business practices. Conversely, when there is a mismatch between the technology and the practices or values of startups, adoption can stall, highlighting the importance of evaluating social media marketing tools for both their value and practical compatibility. Thus, the extent to which social media marketing is compatible with a startup's current operations and goals significantly impacts its adoption and subsequent utilization, highlighting the need for startups to assess how well these innovative marketing tools fit within their existing business ecosystem before full-scale implementation (Bhagat et al., 2021; Effendi et al., 2020; Leeflang et al., 2014).

H1 Technology compatibility has a significant positive impact on social media marketing adoption.

Technology complexity and social media marketing adoption

Navigating through the complications of social media marketing presents a unique set of challenges for businesses, particularly startups, largely due to the complexity of technology involved. The process of leveraging various online platforms and tools to engage with customers and promote products requires not just a basic understanding but an in-depth knowledge of the ever-evolving digital landscape (Febriyantoro, 2020). For many, especially those without the necessary expertise or resources, this complexity can be overwhelming, making it difficult to keep pace with new trends and effectively implement sophisticated social media marketing strategies (Amoah & Jibril, 2020). Moreover, the inherent details and the constant introduction of new features and algorithms in social media platforms intensify this challenge, often leading to a hesitancy in adoption. This reluctance is often further exacerbated by psychological barriers, such as technology anxiety or phobia, in addition to practical concerns, such as managing vast amounts of information, navigating internet security risks, and resolving frequent technical glitches (Chakraborti et al., 2022; Sharma et al., 2020). Therefore, the perceived technological complexity not only imposes a significant hurdle but also has a profound negative impact on the willingness and ability of startups to fully embrace and benefit from social media marketing adoption.

H2 Technology complexity has a significant negative impact on social media marketing adoption.

Technology cost and social media marketing adoption

The cost associated with acquiring and implementing technology, particularly in the context of social media marketing, poses a significant barrier to its adoption among startups. Beyond the initial outlay for hardware and software, startups must account for expenses related to staff training and development, ongoing maintenance, and digital data storage, all of which can accumulate quickly and often exceed initial estimates (Eze et al., 2021; Guzana & Msosa, 2022). Moreover, the opportunity costs, including potential revenue losses during the transition to new technology cannot be overlooked. Despite the anticipated cost reductions technology adoption promises, with some studies suggesting up to 90 percent savings, the reality for startups often reflects a mere fraction of these savings, with only 6 percent experiencing the expected cost reduction (Neubert & Van Der Krogt, 2019). This discrepancy likely results from lack of insights into the full spectrum of costs involved and the tangible benefits social media marketing may deliver. The perception of technology expenditure as a dead investment, coupled with the risk of additional, unforeseen costs, contributes to a reluctance among startups to invest in social media marketing. This hesitancy is further intensified by the financial constraints of startups, making the significant negative impact of technology cost on social media marketing adoption an issue of significant concern for businesses navigating the early stages of growth.

H3 Technology cost has a significant negative impact on social media marketing adoption.

Technology infrastructure and social media marketing adoption

The foundation of a robust technology infrastructure, including both hardware and software components, plays a pivotal role in enabling startups to adopt and excel in social media marketing. In technology-driven landscape, the move towards virtualization and cloud servers not only increases the burden of substantial initial IT setup costs but also provides scalable solutions tailored to the unique nature and demands of the business (IBM, 2022; Ullah & Lai, 2013). This optimal IT framework is instrumental in supporting startups by enhancing their capacity for data management, boosting security and privacy measures, and ensuring uninterrupted operations, that are key factors contributing to an effective social media marketing strategy. Moreover, a well-established technology infrastructure allows startups to move beyond basic digital tools, expanding their digital footprint, reaching a wider audience, and harnessing advanced data analytics to inform and refine their marketing efforts. The limitations imposed by inadequate IT infrastructure, such as reliance on outsourced functions and the issues with costs and quality management, emphasize the importance of a strong technological base. Therefore, investing in and prioritizing a comprehensive technology infrastructure significantly positively impacts ability of startups to adopt, implement, and benefit from social media marketing, setting a solid foundation for competitive advantage and growth in the digital realm (Abed, 2020; Guzana & Msosa, 2022; Ratnasingam et al., 2021).

H4 Technology infrastructure has a significant positive impact on social media marketing adoption.

Technology knowledge and social media marketing adoption

Technology knowledge plays an important role in the successful adoption and utilization of social media marketing by startups. Firms equipped with employees who possess a deep understanding of digital tools and platforms can leverage these technologies more effectively, crafting targeted content, interpreting data analytics accurately, and navigating complex social media landscapes with greater ease (Alami et al., 2022; Prasetya & Susilo, 2022). This proficiency in handling technology not only enhances the capability of businesses to engage with their intended audience through tailored marketing campaigns but also significantly improves engagement and conversion rates, contributing to the overall success of their social media marketing efforts. In contrast, startups with limited technological knowledge may find themselves unable to fully exploit the potential of social media marketing, missing critical opportunities for growth and engagement due to a lack of understanding of the tools available to them (Choi et al., 2010; Venkatesh et al., 2020).

H5 Technology knowledge has a significant positive impact on social media marketing adoption.

Mediating role of technology knowledge

Technology knowledge emerges as a key mediator in the relationship between technology compatibility, complexity, cost, and infrastructure, and social media marketing adoption by startups. Firms equipped with a workforce that possesses a comprehensive understanding of digital tools and platforms are in a better position to navigate the intricacies of social media marketing, despite challenges related to technology complexity and the initial and ongoing costs associated with digital technologies (Alami et al., 2022; Prasetya & Susilo, 2022). This knowledge enables businesses to more effectively integrate social media marketing strategies with their existing business models, ensuring that these technologies complement and enhance their core operations and value proposition (Bizhanova et al., 2019; Petković et al., 2018). Furthermore, well-informed individuals and teams can overcome hurdles related to outdated or inadequate technology infrastructure by identifying and leveraging the most suitable platforms and tools that align with their specific business needs and financial constraints, thus optimizing their social media marketing efforts for maximum impact and efficiency (Alainati, 2022; Aleinikova et al., 2023). Consequently, the presence of technology knowledge within a firm not only facilitates a smoother transition and more effective utilization of social media marketing technologies but also acts as a key factor that enables businesses to overcome potential barriers presented by technology compatibility, complexity, cost, and infrastructure challenges (Fig. 1).

H6 Technology knowledge significantly mediates the relationship between technology compatibility and social media marketing adoption.

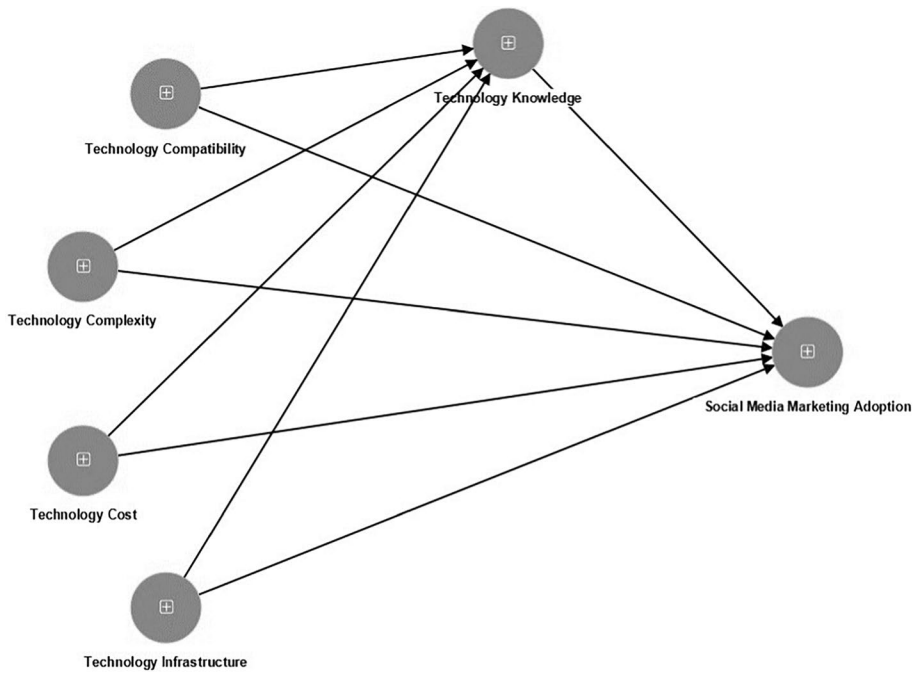


Fig. 1 Conceptual framework (2024) Source: Developed by authors using smart PLS

H7 Technology knowledge significantly mediates the relationship between technology complexity and social media marketing adoption.

H8 Technology knowledge significantly mediates the relationship between technology cost and social media marketing adoption.

H9 Technology knowledge significantly mediates the relationship between technology infrastructure and social media marketing adoption.

Methodology and analysis

This study uses a quantitative research design, focusing on technological determinants of social media marketing adoption among Pakistani SMS. The population consists of SMS defined by the (SMEDA) of Pakistan. The city of Karachi was chosen for its economic significance and representative diversity of Pakistani startups. A sample of 1,000 startups was randomly selected from the list provided by SMEDA. Simple random sampling technique was used to maintain the integrity and representativeness of the sample. The target was to collect responses from at least 384 startup owners to ensure statistical significance and reduce the margin of error (Krejcie & Morgan, 1970). The researchers were successful in getting response from 386 startups.

The survey questionnaire was chosen as the main instrument for data collection due to its direct engagement with respondents and the efficiency in handling large data sets. The questionnaires were distributed electronically via WhatsApp. All questionnaire items in the survey were measured on a five-point Likert scale excluding demographics. The five-point scale offers respondents a wide range of options, granting them the choice to select the 'precise' answer instead of settling for an option that is merely 'close' or 'approximate' (Taherdoost, 2019). The survey was designed to be bilingual (Urdu and English) to accommodate language preferences of respondents, increasing understanding and response accuracy. The primary unit of analysis in this research was the individual startup owners, as they play a crucial role in decision-making processes related to social media marketing strategies. In this study, the consent was obtained from the respondents beforehand. The participation was voluntary and the researcher assured the participants that their data will remain confidential and will be used exclusively for the purposes of this research.

The survey items, excluding demographic questions, were adapted from established literature to ensure content validity. Items related to social media marketing adoption and its influencing factors were adapted from (Bruce et al., 2023). For technological factors (technology complexity, technology knowledge, technology compatibility, technology cost and technology infrastructure) the items were adapted from (Lian et al., 2014; Maduku, 2021), (Bostancıoğlu & Handley, 2018), (Al-Qirim, 2007; Kuo & Lee, 2011), (Lian et al., 2014; Maduku, 2021) and (Jabbouri et al., 2016), respectively.

In our study examining the impact of technological factors on social media marketing adoption, partial least squares–structural equation modeling (PLS–SEM) was used as the analytical tool. PLS–SEM is a multivariate technique that allows for the simultaneous analysis of multiple relationships among observed and latent variables,

providing a comprehensive understanding of the complex interplay between technological determinants and social media marketing adoption. The use of PLS–SEM adds significant value and rigor to our analysis by allowing for a detailed exploration of the mediating effects of technology knowledge and other complex relationships within our conceptual framework. Previous studies have successfully employed PLS–SEM in similar contexts (Sarstedt et al., 2020), establishing and justifying its suitability and robustness in examining the factors influencing social media marketing adoption in this study. In addition, PLS–SEM involves testing both the measurement model and the structural model, ensuring the reliability and validity of the constructs and their relationships.

Following the data collection, the measurement model, as suggested by Henseler et al. (2015) was applied to ensure the reliability and validity of the instrument. Initially, the model was checked for factor loading. As per the recommendation by (Hair et al., 2012), the items with factor loading less than 0.708 were to be excluded. In this research, with the exception of one indicator, TKNOW1 (Indicator to measure technology knowledge; factor loading 0.670), analysis showed that all factor loadings were above this threshold and there were no significant cross loadings. In spite of this, the decision was taken to keep this item in the model in accordance with the arguments by (Hair, 2010) and Yana et al. (2015), according to which factor loadings greater than 0.6 can be considered appropriate for exploratory research. The results are mentioned in Table 1

After ensuring the item loadings, the next step was to check for the validity and reliability of the research instrument followed by discriminant validity. Table 2 indicates that all constructs in the survey, related to social media marketing adoption and technological factors, exceed the threshold values for reliability and validity. Cronbach's alpha and composite reliability scores are all above the accepted threshold of 0.7, confirming the reliability of the survey. The Average Variance Extracted (AVE) values for each construct are also above the minimum threshold of 0.5, confirming the validity of the measures (Hair et al., 2012).

The next step was to check for discriminant validity. Schwab (2013) recommended that discriminant validity is present when different constructs are empirically distinct. For the measurement model of this study, the heterotrait–monotrait (HTMT) ratio of correlations and the Fornell–Larcker criterion were employed to check discriminant validity. For HTMT, Kline (2017) suggested a more conservative threshold of 0.85, whereas Gold et al. (2001) propose a slightly lenient cutoff of 0.90. In this study, the more conservative criterion of 0.85 was adopted. The findings show that the highest HTMT ratio obtained was 0.439, falling below the set threshold and thereby confirming discriminant validity for the constructs examined. The results for discriminant validity are mentioned in Tables 3 and 4, respectively.

After establishing the reliability and validity of the research instrument, structural model was tested to examine the relationships and ascertain the significance of the independent variables selected from the literature review. For the purpose of testing the framework, the gathered data were analyzed using a bootstrapping sample size of 5000 using Smart PLS 4. The dependent variable in the study, Social Media Marketing Adoption, is denoted as (SM), while independent variables are represented

Table 1 Item loadings

Items	Social Media Marketing Adoption	Technology Compatibility	Technology Complexity	Technology Cost	Technology Infrastructure	Technology Knowledge
SM1	0.833					
SM2	0.872					
SM3	0.824					
SM4	0.902					
SM5	0.892					
SM6	0.908					
SM7	0.870					
TCOM1		0.869				
TCOM2		0.917				
TCOM3		0.917				
TCOM4		0.888				
TCOMX1			0.840			
TCOMX2			0.760			
TCOMX3			0.737			
TCOMX4			0.710			
TCOST1				0.892		
TCOST2				0.921		
TCOST3				0.922		
TCOST4				0.904		
TINF1					0.799	
TINF2					0.890	
TINF3					0.872	
TINF4					0.892	
TINF5					0.890	
TKNOW1						0.670
TKNOW2						0.831
TKNOW3						0.814
TKNOW4						0.842

Source: Authors' work, 2024

Table 2 Reliability and validity

Variables	Cronbach's alpha	Composite reliability	Average variance extracted (AVE)
Social media marketing adoption	0.947	0.947	0.760
Technology compatibility	0.919	0.922	0.806
Technology complexity	0.783	0.904	0.579
Technology cost	0.931	0.933	0.828
Technology infrastructure	0.919	0.926	0.756
Technology knowledge	0.800	0.813	0.627

Source: Authors' work, 2024

as technology compatibility (TCOM), technology complexity (TCOMX), technology cost (TCOST), technology infrastructure (TINF) and technology knowledge (TKNOW). The results of the hypotheses testing are mentioned in Table 5.

Table 3 Discriminant validity heterotrait–monotrait ratio (HTMT)

Variables	SM	TCOM	TCOMX	TCOST	TINF	TKNOW
Social media marketing adoption						
Technology compatibility	0.439					
Technology complexity	0.142	0.215				
Technology cost	0.172	0.323	0.165			
Technology infrastructure	0.377	0.559	0.168	0.285		
Technology knowledge	0.478	0.519	0.215	0.188	0.413	

Source: Authors' work, 2024

Table 4 Discriminant validity Fornell–Larcker criterion

Variables	SM	TCOM	TCOMX	TCOST	TINF	TKNOW
Social media marketing adoption	0.872					
Technology compatibility	0.412	0.898				
Technology complexity	0.150	0.201	0.761			
Technology cost	0.162	0.298	0.139	0.910		
Technology infrastructure	0.362	0.516	0.165	0.265	0.870	
Technology knowledge	0.422	0.447	0.194	0.163	0.358	0.792

Source: Authors' work, 2024

Table 5 Null hypotheses

Hypothesis	Relationships	T statistics (|O/STDEV|)	P values	Result
1	TCOM does not have a significant positive impact on SMM	2.860	0.004	Not supported
2	TCOMX does not have a significant negative impact on SMM	0.541	0.588	Supported
3	TCOST does not have a significant negative impact on SMM	0.253	0.801	Supported
4	TINF does not have a significant positive impact on SMM	2.866	0.004	Not supported
5	TKNOW does not have a significant positive impact on SMM	4.585	0.000	Not supported

Source: Authors' work, 2024

Our H1 asserts that Technology Compatibility has a significant positive impact on Social Media Marketing Adoption. The findings support the alternative hypothesis and reject the null hypothesis, with a beta coefficient of 0.203 and a t-statistic of 2.860 and a p value of 0.004. This confirms that startups with higher compatibility with existing technologies are indeed more inclined towards social media marketing adoption. H2 proposes that Technology Complexity has a significant negative impact on Social Media Marketing Adoption. The results do not support this hypothesis; hence our null hypothesis is supported. With a beta coefficient of 0.029 and a t-statistic of 0.541, and a p value of 0.588, it appears that the complexity of technology does not significantly prevent startups from social media marketing adoption. H3 in our research suggests that Technology Cost has a significant negative impact on Social Media Marketing Adoption. Similar to H2, the study's findings do not validate this

hypothesis, and hence the null hypothesis is supported. The beta coefficient is 0.013, the t-statistic is 0.253, and the *p* value is 0.801, indicating that the cost of technology is not a significant barrier to the adoption of social media marketing within the examined startups. For H4, that Technology Infrastructure has a significant positive impact on Social Media Marketing Adoption, the statistical analysis with a beta coefficient and a t-statistic of 0.153 and a *p* value of 0.004 substantiates this hypothesis and rejects the null hypothesis, highlighting the importance of having adequate technology infrastructure for the social media marketing adoption. For H5, that Technology Knowledge has a significant positive impact on Social Media Marketing Adoption, the alternative hypothesis is strongly supported and null hypothesis is rejected, with a beta coefficient of 0.270 and a t-statistic of 4.585, and the *p* value is less than 0.001. This demonstrates that extensive knowledge of technology is a powerful facilitator for the effective adoption of social media marketing in Pakistani startups.

After examining all the relationships, another important aspect that significantly influences social media marketing adoption, technology knowledge, was introduced as a mediator in the model. The findings of the moderated model are presented in Table 6.

Introducing technology knowledge as a mediator revealed that the presence of Technology Knowledge as a mediator strengthened the positive relationship between Technology Compatibility and Social Media Marketing Adoption (H6). This implies that startups with comprehensive technology knowledge are more proficient at selecting and adopting technologies that align well with their business strategies, particularly in the context of social media marketing. For H7, when Technology Knowledge is introduced as a mediator, businesses have a better chance of navigating and understanding complex technology. This knowledge reduces the challenges associated with complexity, making it less of an obstacle in social media marketing adoption. Essentially, knowledge empowers businesses to overcome the barriers that complexity could present. For H8, the study found that having a better understanding of technology does not significantly change the impact that the technology cost has on social media marketing adoption. Even if a business has technology knowledge, the technology cost seems to be a distinct factor in the decision-making process, unaffected by how much knowledge business have about the technology. For H9, Technology Knowledge strengthened the positive effect of Technology Infrastructure on Social Media Marketing Adoption, supporting the idea that understanding the infrastructure can lead to better utilization and adoption social media marketing.

Table 6 Mediating effects

Hypothesis	Relationships	Original sample (O)	Sample mean (M)	T statistics (|O/STDEV|)	*p* values	Result
H6	TCOM—> TKNOW—> SM	0.092	0.095	3.048	0.002	Supported
H7	TCOMX—> TKNOW—> SM	0.026	0.028	1.994	0.047	Supported
H8	TCOST—> TKNOW—> SM	0.001	0.001	0.075	0.940	Not Supported
H9	TINF—> TKNOW—> SM	0.044	0.046	2.357	0.018	Supported

Source: Authors' work, 2024

After examining the direct and moderating effects, the Stone–Geisser test (Q^2) was used to assess the predictive relevance of the model. If Q^2 is greater than zero for a particular endogenous construct, the model has predictive relevance (Henseler et al., 2015). Our Q^2 value (0.177) was greater than zero, hence confirming the predictive relevance of the model. In addition, the overall fit of the model was also assessed using standardized root mean residual index (SRMR) and Normed fitness index (NFI). Ringle et al. (2020) suggested a standardized root mean square residual (SRMR) value below 0.10 for a satisfactory fit for the overall model. In our case, the value of SRMR is 0.051, claiming that the model studied has a perfect fit. For the Normed Fit Index (NFI), Kline (2017) recommends that it should fall within the 0 to 1 range, with values above 0.9 indicating a good fit. The NFI value for the model studied is 0.913, confirming the model's good fit according to this index. The results of overall model fit are presented in Table 7.

Discussion

The findings highlight the critical role of technology compatibility in social media marketing adoption by startups. It emphasizes that startups prefer social media marketing tools that align with their existing operational frameworks, primarily because such compatibility enhances resource optimization, maintains operational continuity, and simplifies the change management process. The seamless integration of compatible technologies ensures that the adoption of new social media tools align well with the business model and does not disrupt ongoing business processes. Furthermore, technology compatibility with the business model reduces resistance to change, and enables startups to quickly see a return on their investments in social media marketing. This observation is consistent with studies conducted by Ullah et al. (2023) on Pakistani SMEs and Effendi et al. (2020) on Indonesian SMEs, which both highlighted the significant influence of technology compatibility on the effectiveness of social media marketing strategies. This consistency across studies suggests a universal principle: the adoption and successful implementation of social media marketing tools are greatly facilitated by their compatibility with the operational framework of the business.

The findings that technology complexity positively impacts the social media marketing adoption in startups, highlights an evolving landscape within the Pakistani startup ecosystem. The absence of technology complexity as a perceived barrier to social media marketing adoption could be linked to several factors. Modern marketing tools are increasingly designed for user-friendliness, easing the integration process. In addition, a cultural shift towards embracing digital innovation within these startups might be underway due to the presence of a tech-savvy workforce expert at tackling complex technological landscapes. This finding is in complete contrast with research conducted in other regions, such as the studies by Ali Abbasi et al. (2022) in Malaysia and Effendi

Table 7 Model fitness indexes

	SRMR	NFI
Acceptable values	≤ 0.10	≥ 0.9
Calculated values	0.051	0.913

Source: Authors' work, 2024

et al. (2020) in Indonesia, which identified technological complexity as a significant hurdle for traditional SMEs in social media marketing adoption. The discrepancy between these findings and the lack of evidence in the Pakistani context might highlight a generational change in technology adaptability. Traditional SMEs often struggle with outdated systems and a workforce that may lack the necessary skills and agility to cope with rapid technological shifts. Conversely, the current generation of Pakistani startups, possibly led by younger, tech-savvy entrepreneurs, seems more equipped and willing to embrace and leverage the complexities of the latest technologies for social media marketing, suggesting a significant paradigm shift in the approach towards technology adoption within the startup domain.

The findings suggest a shift in how startups perceive investment in technology. This shift suggests that the financial aspect of social media marketing adoption does not pose a significant barrier for startups. One explanation for this could be the availability of cost-effective, and even free social media marketing tools. These tools provide startups with the flexibility to scale their social media marketing campaigns in alignment with their growth and financial capacity, to avoid over spending on initial investments. This understanding contrasts with the findings from prior researches, such as the studies by Ahmad Tarmizi et al. (2020), Effendi et al. (2020), Guzana and Msosa (2022) and Ullah et al. (2023), which highlighted technology cost as a significant hurdle in social media marketing for SMEs. Traditional SMEs operate in established and competitive markets and often exhibit a cautious approach towards new financial commitments, and prioritize immediate and tangible returns due to budget constraints and risk aversive nature. In contrast, startups approach social media marketing investments from a strategic perspective, focusing on the long-term benefits and competitive edge that such technologies can provide. Startups are more inclined to invest in social media marketing as they are driven by the urgency to establish a market presence and expand their customer base. They may view the costs associated with social media marketing adoption as an integral part of their growth strategy and innovation-driven mindset, rather than only a prohibitive expense.

The findings assert that technology infrastructure positively influences social media marketing adoption, emphasizes on the critical need for startups to invest in technological foundations to thrive in the digital marketing domain. This infrastructure, including reliable internet access and the latest hardware and software, is pivotal for seamless integration of various social media marketing tools. Such a foundation not only facilitates smoother operations but also enhances startups' ability to engage effectively with their audience across diverse digital platforms, thereby unlocking the full potential of social media marketing tools. This finding aligns with the findings from prior studies by Islam et al. (2021), Miklosik et al. (2019) and Ratnasingam et al. (2021), which collectively support the essential role of technology infrastructure in promoting social media marketing adoption among SMEs. This similarity between the strategies of startups and SMEs demonstrates a shared understanding of the importance of technological infrastructure in social media marketing success.

The findings regarding the role of technology knowledge in facilitating social media marketing adoption strategies within startups suggests that having a well-informed team is key to driving marketing innovation. Startups equipped with technologically

knowledgeable staff or those that invest in training of employees are in a better position to leverage the full benefits offered by social media marketing. Similarly, literature focusing on conventional SMEs highlights a significant gap in social media marketing knowledge, with studies by Beier and Wagner (2016), Lee and Tao (2018), and Ratnasingam et al. (2021) identifying this lack of expertise as a major obstacle to the adoption and effective implementation of social media marketing strategies. While both startups and SMEs face challenges due to a lack of expertise, SMEs appear to struggle even more with social media marketing adoption as they often encounter difficulties due to their established operational structures resulting in potential resistance to change.

The significant mediating role of technology knowledge in the relationships between technology compatibility, complexity, and infrastructure, and social media marketing adoption, suggests how thorough understanding and expertise in social media marketing can significantly benefit startups. This expertise could enable more effective selection and adoption of technologies that are in accordance with their business strategies (Backfisch et al., 2020; Tsou & Chen, 2023).With technology knowledge, businesses are better positioned to navigate and comprehend complex technological landscapes. This understanding and knowledge, not only simplifies the adoption process, it also ensures that the chosen technologies are optimally utilized within their social media marketing efforts. Essentially, technology knowledge serves as a critical bridge, making the integration of social media marketing tools more intuitive and aligned with the operational needs and goals of the company (Allioui & Mourdi, 2023). This capability to proficiently manage technology-related decisions highlights the importance of investing in technology knowledge as a core competency within startups, promoting a more strategic and effective adoption of social media marketing practices.

Theoretical and practical implications

This research provides theoretical implications by analyzing how technological factors of compatibility, complexity, cost, infrastructure, and knowledge interact to influence social media marketing adoption in SMS. The findings reveal that technology knowledge plays an important mediating role, enabling startups to effectively navigate technological complexities and align compatibility and infrastructure for social media marketing adoption. This suggests a shift in the theoretical understanding of technology adoption, highlighting the necessity of incorporating technology knowledge into models of social media marketing adoption. The study points towards a generational change in how SMS approach technology challenges in contrast to traditional SMEs, and highlights the distinct and unique nature of these newly incorporated firms, hence necessitating the need of studying these businesses separately rather than merging their studies with traditional SMEs.

From practical/managerial perspective, the research offers valuable insights for startups aiming to optimize their social media marketing strategies. First, the study emphasizes on the importance of investing in technology knowledge and training within the organization. This is not just for leveraging existing technologies but also for effectively navigating the complexities and integrating new technologies that align with the strategic goals of startups. Second, startups should invest in social media tools that can easily integrate with current systems and those that will continue to support their evolving

business needs rather than being concerned by costs and technological challenges. This strategic approach allows them to use their resources wisely, adopting social media marketing tools that offer the best return on investment and facilitate their business objectives. Third, as startups are not deterred by technological challenges, government should launch a Digital Startup Support Initiative for providing subsidized access to premium social media marketing tools. For boosting technology knowledge, initiatives related to social media marketing training and workshops should be introduced.

Limitations and future research

This study, while providing insights into technological factors for social media marketing adoption among Karachi based startups, has several limitations. Geographically, the study solely focused on Karachi which may not reflect the discrepancies across diverse cities of Pakistan, that as a consequence, may limit the generalizability of the results. In addition, the majority of participants in this study were under the age of 25. This may bias the insights towards perception of young entrepreneurs. Finally, the technological factors studied in this research were derived from research on established SMEs and may not fully align with the distinct environment and challenges faced by startups.

For future research, it would be beneficial to broaden the study by including startups from other cities of Pakistan. Including entrepreneurs of various ages would also give a comprehensive understanding of how digital marketing is adopted and perceived as across generations. The future studies should focus on factors that are specific to startups, instead of applying factors related to SMEs. This could be done through conducting an exploratory study with startups. In addition, the researchers should explore the impact of other variables, such as cultural influences or market maturity, to further clarify the complexities of social media marketing adoption in startups.

Conclusion

SMM stands as an important technology for growth of the business as it enhances visibility, engages directly with customers, and promotes brand loyalty. Particularly for SMS, SMM offers a cost-effective medium to compete in the global marketplace, reach a wider audience, and drive growth. Keeping in view, the study focused on the technological determinants responsible for SMM adoption with technology compatibility, infrastructure and knowledge showing a direct significant impact. Furthermore, startups' technology adoption was not impacted by the complexity and cost associated with it, highlight their unique nature of dealing with technology as compared to SMEs. The significant mediating role of technology knowledge suggests that understanding and expertise in SMM can significantly benefit startups in dealing with technological barriers. The research highlights the need for tailored strategies and support systems to navigate technological challenges. It also suggests targeted investment in technology training and infrastructure to empower startups with the tools and skills necessary for leveraging social media marketing effectively, promoting innovation and growth in the digital age.

Abbreviations

SMS	Small and medium startups
SMEDA	Small and medium development authority
SME	Small and medium enterprises

SMM Social media marketing

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Author contributions
AA was primarily responsible for conceptualizing the research study, its scope, methodology, data collection, analysis and writing of the overall draft. SR added her valuable insights in the conceptualization of the model and overall designing and flow of the manuscript. She also contributed towards journal identification. NMS provided guidance and oversight throughout the research process by reviewing and providing valuable feedback on the methodology, data analysis, and manuscript drafts. MWR added his insights after the review process and transform the manuscript as per reviewers' comments. IAK helped with the methodology part and final review and proofreading of the manuscript. ZA contributed by addressing reviewers' concerns and proofreading of final manuscript. Final manuscript was also proofread by all the authors.

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Declarations

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